



Introduction – Networking & Resumes

Workshop 1

Club Mission

The Financial Quants and Engineers (FQE) club will develop students' technical skills and network at the intersection of quantitative finance and technology. Members will work on a variety of projects within data science, risk analysis, quantitative modeling, and algorithmic trading. The main goal of the organization is to provide the necessary exposure and skills for students to gain roles in quantitative fields.

FQE's Place in Baruch

1. FQE intends to create a community of like-minded undergraduate students at Baruch to learn more about the quantitative finance industry
2. We do this by:
 - Leveraging current resources available at Baruch College
 - Posing challenging problems to our members with real world implications through relevant workshops and education sessions
 - Challenge members with developing their skills and building their portfolios through hands-on projects
 - Build a strong network of professionals and alumni in the quantitative finance industry
3. Regardless of current ability, we aim to promote skills around:

○ Asset Pricing	○ Probability Theory	○ Data Science
○ Risk Management	○ Statistics	○ Machine Learning/AI
○ Portfolio Optimization	○ Differential Equations	○ Analytical & Critical Thinking
○ Algorithmic and High-Frequency Trading	○ Linear Algebra	○ Problem-Solving
○ Quantitative Investing	○ Numerical Methods	○ Communication
○ Stochastic Calculus	○ Programming (C++/Python)	○ Attention to Detail

FQE Board

President



Arthur Faynin

Vice-President



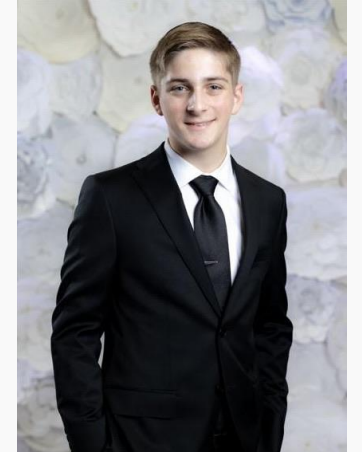
Alfred Leong

Treasurer



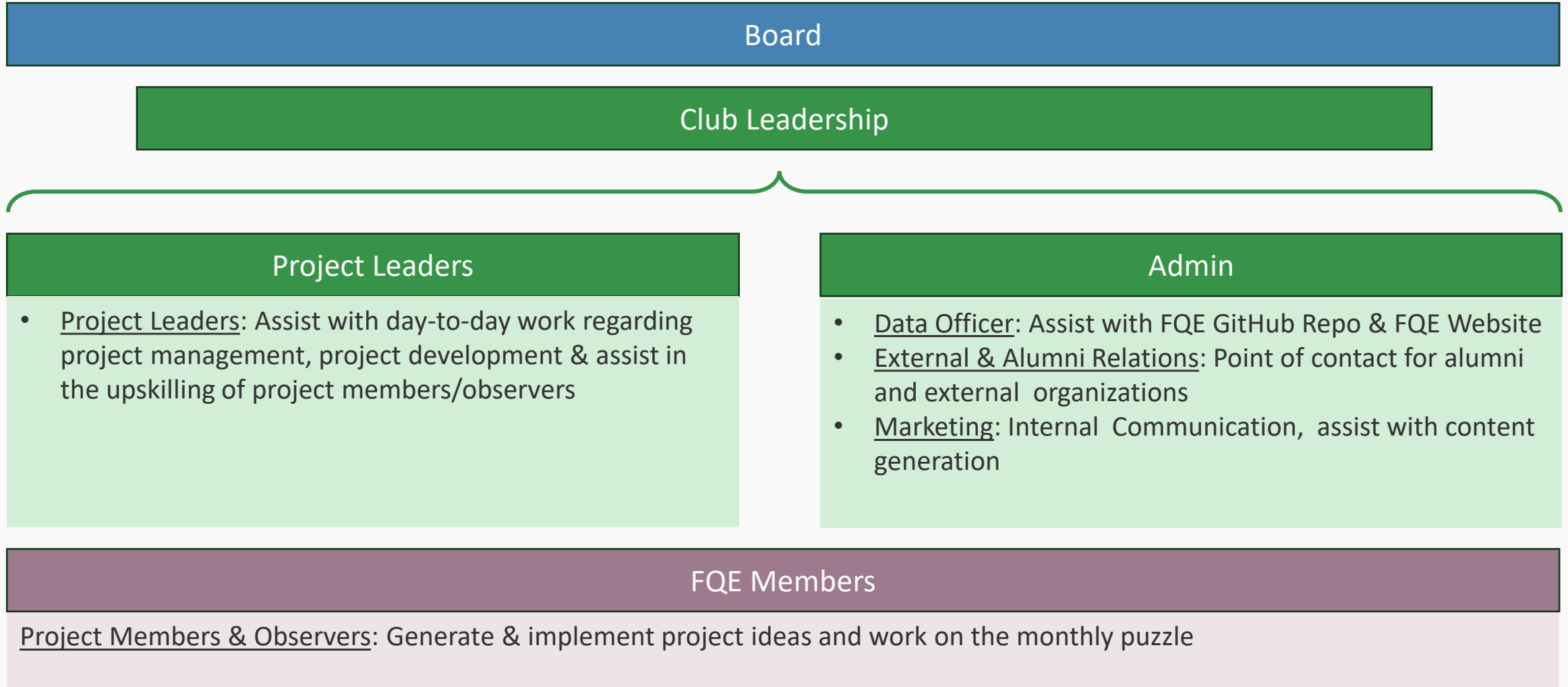
Alexi Tilly

Secretary



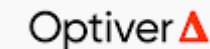
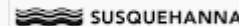
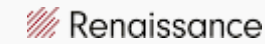
Abie Lent

Club Structure



Workshop Overview

- Brainteaser & Internal Networking
- Careers in Quantitative Finance
- Firms hiring in Quantitative Finance
- Networking & Resumes
- Resources Moving Forward
- How this all relates to the project teams



Expectations

- Design

- Workshop structure, resume review, mock interviews, guest speakers
- Designed to introduce fundamentals to quant finance through a collaborative environment and project teams

- Objectives

- Complete projects by end of semester
- Foster quantitative skill development & community
- Create a fun and collaborative environment

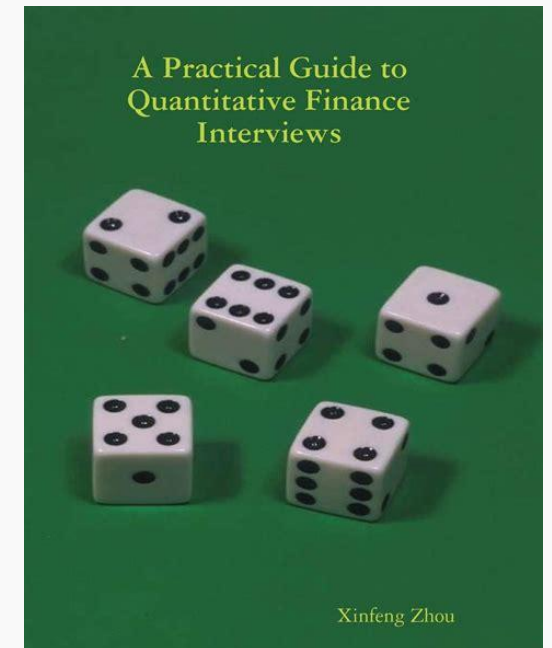
- Expectations

- Instilling best practices – communicate, arrive on time, step outside to use your phone, etc.
- Come prepared
- Utilize resources and reach out to leaders for guidance with questions
- Personal effort dictates your likelihood to end up in a career in quant finance

Brain Teaser & Networking

- Tigers & Sheep

- One hundred tigers and one sheep are put on a magic island that only has grass. Tigers can eat grass, but they would rather eat sheep. Assume:
 - A. Each time only one tiger can eat one sheep, and that tiger itself will become a sheep after it eats the sheep.
 - B. All tigers are smart and perfectly rational and they want to survive.
- So will the sheep be eaten?



Project Interest Form, Discord, & Linked-in Group

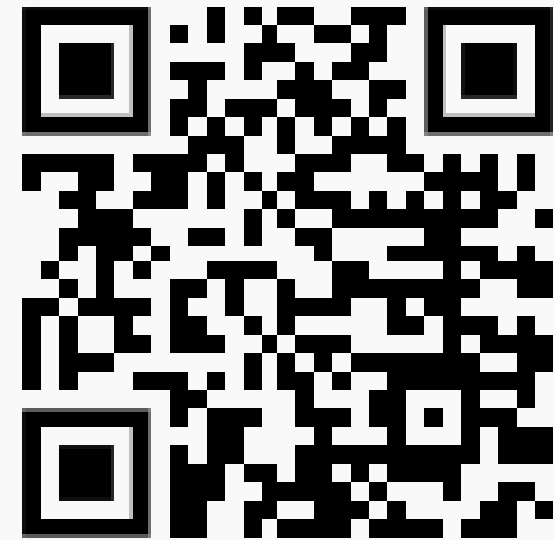
Project Interest Form: <https://forms.gle/bT16nC2aDwgB6zeR8>

Join our LinkedIn Group: <https://www.linkedin.com/groups/14991054>

Project Interest Form



LinkedIn Group



Brain Teaser & Networking

- Tigers & Sheep (*answer*)

- So, will the sheep be eaten?

- 100 is a large number, so again let's start with a simplified version of the problem.

- If there is only 1 tiger ($n = 1$), surely it will eat the sheep since it does not need to worry about being eaten.

- How about 2 tigers?

- Since both tigers are perfectly rational, either tiger probably would do some thinking as to what will happen if it eats the sheep. Either tiger is probably thinking: if I eat the sheep, I will become a sheep; and then I will be eaten by the other tiger. So, to guarantee the highest likelihood of survival, neither tiger will eat the sheep.

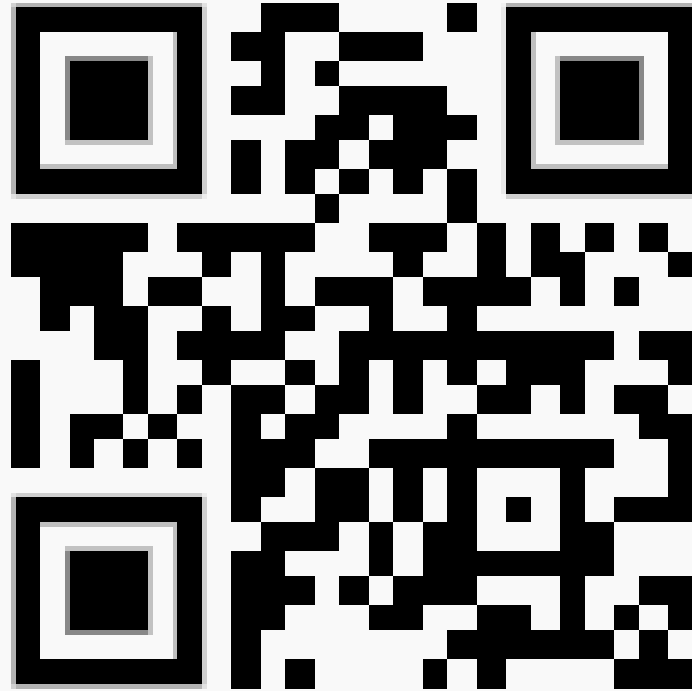
- If there are 3 tigers, the sheep will be eaten since each tiger will realize that once it changes to a sheep, there will be 2 tigers left, and it will not be eaten. So, the first tiger that thinks this through will eat the sheep.

- If there are 4 tigers, each tiger will understand that if it eats the sheep, it will turn to a sheep. Since there are 3 other tigers, it will be eaten. So, to guarantee the highest likelihood of survival, no tiger will eat the sheep.

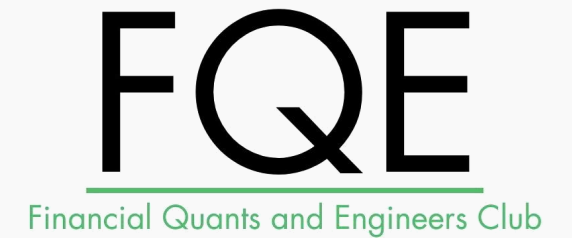
- Following the same logic, we can naturally show that if the number of tigers is even, the sheep will not be eaten. If the number is odd, the sheep will be eaten. For the case $n=100$, the sheep will not be eaten.

Discord Link

Discord Link: <https://discord.gg/GKaQkhMzg>



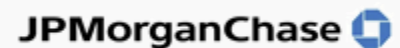
Careers & Firms in Quant Finance



Careers in Quant Finance (1 of 7)

- Quantitative Research & Analysis (Quant Research)

- Professionals in this area use statistical and quantitative methods to analyze and predict the markets, and apply programming tools to produce robust investment strategies. Their work revolves around creating mathematical models that are used to assess and manage financial systems, potential risk, and timing of trades.
- Necessary Skills: a strong command of programming languages, such as Python, C#, and SQL, as well as statistical analysis tools, such as R, Matlab, and SAS. Some roles will also require knowledge of machine learning and natural language processing techniques. Good understanding of a variety of mathematical and statistical models used in finance.



Careers in Quant Finance (2 of 7)

- Portfolio Management

- Portfolio managers engage in portfolio construction, monitoring asset exposures and allocations, managing client requests, tax management, monitoring pre-trade client guideline compliance and exception resolution. They initiate trades, and monitor the portfolios on an ongoing basis. They also develop a deep understanding of investment products and operational policies and procedures. With career progression, they can manage a team of analysts and researchers.
- Necessary Skills: in addition to effective communication skills and knowledge of asset classes, professionals in this area also require strong quantitative and mathematical modeling, coding, and analytical thinking skills. This role often prefer a financial analyst certification, like the Chartered Financial Analyst (CFA), and previous experience. Most portfolio managers will start their careers as portfolio analysts.



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P I M C O



Careers in Quant Finance (3 of 7)

- Programming & Software Development (Quant Dev)

- Quantitative engineers or quantitative developers work in the FinTech space. They are responsible for designing, developing, testing and deploying sophisticated software solutions to facilitate the work of various financial institutions.

- Necessary Skills: excellent coding skills in Python, C++, and Java, and knowledge in probability, linear regression and time series data analysis. In addition, interest in financial markets and knowledge of various financial products give quant developers a distinct advantage, since they work on a variety of projects with teams across an organization.

BlackRock

 **enfusion**

 **hc** technologies


SERRALA

Careers in Quant Finance (4 of 7)

- Risk Management

- Professionals in this area empower the decision making process for investments and trades by providing risk analysis, and developing/enhancing risk model frameworks across various markets and assets. They use various techniques, including "value at risk" (VaR), Monte Carlo simulation, and linear regression-based statistical models, to measure the potential of loss on an investment profile. They also run stress tests to gauge the effectiveness of their models.
- Necessary Skills: strong skills in communication and detail orientation, quantitative and financial modeling skills, programming abilities using tools like VBA, Python, R, and SAS, as well as knowledge of various statistical and volatility models.



Careers in Quant Finance (5 of 7)

• Trading (Quant Trading)

- Traders analyze market data, such as price and volume, and use mathematical and statistical models to identify and execute trading decisions that may involve hundreds of thousands of shares and securities.
- A trader develops a strategy and applies the model to historical market data so that it can be back-tested and optimized. If the strategy yields profit, it is then applied onto real-time markets to implement an automated trading process. Quantitative trading techniques also include high-frequency trading, algorithmic trading and statistical arbitrage.
- Necessary Skills: a strong background in programming skills in Python, C++, SQL, R, and/ or Java. Ability to navigate price indexes, such as SPX and VIX. It also requires the knowledge of statistical analysis, numerical linear algebra, and machine learning processes. In addition, traders must possess the ability to thrive under pressure, maintain focus despite long hours, withstand an often competitive/intense environment, and respond well to failure.

BELVEDERE
TRADING

CTC[®]
CHICAGO TRADING COMPANY[®]

imc

Morgan Stanley

sunrise Futures

Careers in Quant Finance (6 of 7)

• Data Science & Analytics

- As financial institutions further integrate the practice of collecting and analyzing data to gauge profit, loss, and client satisfaction, data science continues to be the fastest growing area of quantitative finance.
- Professionals in this area work on data mining, gathering data sets, and deriving insights from these data sets. Data Scientists work in many data driven companies, such as investment banks, asset management firms, and technology companies. Their roles typically focus on risk management and predictive analytics. Data Scientists are increasingly using machine learning, clustering algorithms, and artificial intelligence to identify unusual data patterns.
- Necessary Skills: command of programming languages used in statistical modeling, such as Python and R, ability to work with large sets of financial data, and strong quantitative analysis skills. Time Series Analysis is also key to analyzing financial data. Machine learning and AI re also areas of growing importance in this field.

Bloomberg

BMO  Harris Bank

EY

McKinsey
& Company

pwc 

Careers in Quant Finance (7 of 7)

- Other Career Fields

–*Consulting*, where responsibilities may include: performing asset and derivative valuations, running Monte Carlo simulations to predict risk in different assets, and deploying statistical modeling and optimization techniques to improve risk management decision making at consulting firms like:



–*Equity Research*, where responsibilities may include: performing valuations on a wide range of illiquid investments, constructing possible outcomes for both short and medium term equity moving events, and maintaining a database of option pricing inputs at companies like:



Firms in Quant Finance

- Firms looking for quantitative talent:
 - Sell-Side Firms:
 - Investment banks like BAML, Citi, Goldman Sachs, create financial products and sell them to institutions, retail investors and to one another
 - Buy-Side Firms:
 - Institutional investors like pension funds and insurance companies
 - Other asset managers
 - Hedge funds/Private Equity firms
 - Exchanges & Clearing Houses
 - Market makers, Citadel
 - Exchanges, NYSE, NASDAQ, ICE, CME
 - Data Providers & Software Vendors
 - Bloomberg, FactSet, CapIQ
 - Regulators
 - Rating & Consulting
 - Fintech

Quantitative finance as a discipline emerged in the 1980s. It is also called financial engineering, financial mathematics, mathematical finance, or ..., computational finance. It uses the tools of mathematics, statistics, and computer science to solve problems in finance. Computational methods have become an indispensable part of the finance industry. Originally, mathematical modeling played the dominant role in computational finance. Although this continues to be important, in recent years data science and machine learning have become more prominent. Persons working in the finance industry using mathematics, statistics and computer science have come to be known as quants.

Initially relegated to peripheral roles in finance firms, quants have now taken center stage. No longer do traders make decisions based solely on instinct. Top traders rely on sophisticated mathematical models, together with analysis of the current economic and financial landscape.... Banks are eager to hire “quantitative traders” who know or are prepared to learn this craft.

Networking & Resume



Networking Tips in Brief

- Be polite and use professional language and correct grammar
 - DO NOT refer to a contact's firm as "you guys" instead use "your firm, company, bank, team, group, etc."
- Spread your net as wide as possible
 - Contacts outside of the groups you are interested in can put you in touch with their colleagues in the groups you are interested in
- Be flexible to your contact's schedule
 - Be prepared to reschedule, especially if that contact is staffed on multiple deals
 - Don't take cancelled and rescheduled sessions personally, continue reaching out
- Ask intelligent questions; avoid bad ones
 - Good questions: deal experience; reasons for choosing firm / industry / group; bank culture / organizational structure; characteristics of a good analyst / summer analyst; recruiting process & timeline; staffing composition
 - Bad questions: anything related to compensation; what does an analyst's typical day look like (by the time you hop on a call you should know this); anything that can be found on the first 3 pages of a google search; "can you get me an interview / job offer"

The Steps to a Successful Networking Call

1. The Email
2. Confirmation of Call
3. The Call
4. The Follow-Up

The Email

Hi XXX,

My name is X. I am a [GRADE LEVEL] at Baruch College studying [MAJOR].

I am interested in pursuing a career in [THEIR FIELD] and am trying to learn more about the industry — I am particularly interested in [FIRM].

{OPTIONAL:#XXX recommended that I reach out to you#}. If you have 15 minutes in the next few weeks, I would love to hear about your career path and your time at [FIRM].

I can be reached at XXX-XXX-XXX or at this email. I have attached my resume for your reference.

All the best,

XXX

The Confirmation of Call [With Agenda]

Hi XXX,

I am looking forward to talking at 11 AM this morning. My agenda for the call is to learn about your background (and share mine), to hear about your time at XXX, and to listen to any advice you may have for someone interested in quant finance.

Best,

XXX

Hi XXX!

I am looking forward to chatting with you today at 3:30 PM.

My agenda for the call is to spend a few minutes introducing ourselves, to hear about your time at Jane Street, and finally to listen to any advice you have for my upcoming summer/recruiting season.

Talk soon!

XXX

The Call

1. You call them! Make sure to have gotten their phone number in a prior communication
2. Say hello, begin the call, lead the beginning of the conversation
3. Introduce yourself – give a 1-minute intro on your background, why you reached out/interested in their field, why you're calling
4. Ask about them...
 - Don't ask obvious questions (what is your field, what does a typical day look like, etc)
 - The more specific the question, the better
5. End the call asking for advice or for other people they can connect you with

REMARK: Always have a strong answer as to why you attended Baruch

Following-Up

Send a follow up within 24 hrs

Hi XXX,

Thank you for taking the time to talk to me yesterday.

I am very impressed by Credit Suisse: it sounds like CS really promotes women in the workplace and facilitates a myriad of opportunities for growth (both in the U.S. and abroad).

After talking to you, I am even more interested in sales and trading - the combination of qualitative and quantitative aspects of the job is intriguing and exciting.

I am really excited about the potential opportunity to come to New York and be a part of the Women's Mentor Program. Thanks for the outline on what the program looks like - I will be submitting my application this evening, and will reach out to your colleague Erin tomorrow morning.

I appreciate how candid and honest you were yesterday. I have re-attached my resume for reference.

Thanks again!

Notes about FQE interviews & resumes

- Granted this is a club interview so it's a lot more relaxed, however:
 - Not a lot of follow up emails, thanking us for our time
 - Not very strong answers to why you're applying, what you want to get out of this, etc.
 - Vague and unprofessional answers
 - Resumes were generally not up to par or disorganized
 - Non-professional behavior, atmosphere, or attire
 - Technical interviews were a hit or miss

- Interviewing is a skill like anything else, everyone can improve one way or another

A Bad Resume

- Interviewers take 10 seconds or less per resume
- Career Objectives suck and don't mean anything, don't include it
- Too much meaningless space takes away from explaining about project or work experience
- Not tailored to the specific job
- Specific bullet points are vague and not actionable
- Confusing and not chronological
- Things are not spelled correctly
- Not ATS friendly
- Vague or non-relevant skills
- No non-professional interests
- No personal information



A Good Resume

- Readable, aesthetically pleasing
- Only one page
- Points that are relevant and important
- No lengthy paragraphs
- Email address has professional image
- Work experience includes non paid opportunities as well
- Bullet points follow format: Action through hard skill lead to impact
- Spell checked and physically reviewed: asking these questions
 1. Will a recruiter find my resume easy to read and impactful
 2. Is the content concise and effective
 3. Is the application a good reflection of me and my experience?
- Refer to the template in the discord and resources

FIRST & LAST NAME	
EMAIL ADDRESS (111) 111-1111 LINKED-IN OPTIONAL	
EDUCATION	
University of California, Riverside	Riverside, CA
BS in Business Administration, Finance (GPA: 3.5)	Graduation Date: June 2017
Organizations/Awards: Honors Program, ALPFA, ASUCR, Dean's Honors List, Chancellor's Honors List, Cum Laude	
WORK EXPERIENCE	
Snap, Inc (Snapchat)	Santa Monica, CA
Operations Specialist – Product Ops	August 2017 – Present
• Research user trends to implement Go-To-Market strategies for a new update called Cheetah, improving user experience and gathering data from queues while increasing customer operations satisfaction rating by 11%. • Developing training materials and operational workflows for macro language processes, creating 10+ infographics and webinars called Shopitors specifically for sign-up issues, password resets, and account conflict queues. • Create and improve operations processes for user inquiries, solving 300+ customer inquiries weekly through CRM platforms including Zendesk, Confluence, and Jira.	
Goodwin's Organics	Riverside, CA
Marketing & Strategy Intern	January 2017 – June 2017
• Generated reports on Excel and PowerPoint to conduct trend analysis and research on competitors, identifying target markets to help optimize the website and introduce stamp cards which leveraged customer retention by 15% YTD. • Collaborated with senior management including marketing director to raise awareness Yoga Classes to 5,000+ followers on social media, using direct marketing techniques to gain 250+ followers weekly on Facebook and Instagram. • Strategized ideas enter a niche market for organic products and Yoga classes, creating new waiver forms and building improvements in collaboration with UC Riverside instructors to increase attendance by 15% each class.	
Kohl's	Los Angeles, CA
Operations Management Intern	June 2016 – August 2016
• Analyzed financial statements, business summary reports, and dashboard reports to attain data based on product comparable percentages and penetration rates for district managers, increasing store efficiency of all employees by 16%. • Collaborated with senior management on multiple functions including customer success and finance, increasing survey submissions by 10% by identifying the painpoints of all customer feedback. • Trained 100+ employees in 6 departments by engaging employees with the company's culture, assessing their strengths, opportunities, and overall skills which improved productivity to do tasks by 43% daily.	
Shervin – Williams	Huntington Beach, CA
Operations/Finance Intern	May 2015 – August 2015
• Presented and implemented recommendations like Employee BBQ, Contractor Pro Show, and Employee Cheat Sheet to senior management for intern project, incorporating strategies with 24 interns which ranked #1 in Western Region. • Reviewed P&L monthly results with 3 managers to identify highest and lowest penetration rates on all products, analyzing all points in each department which increased efficiency of deliveries to store by 17% each month. • Maintained and built relationships with 100+ DIY customers and contractors, providing exceptional customer service and efficient paint deliveries daily which translated to exceeding sales goals by 11% MTD.	
LEADERSHIP EXPERIENCE	
Association of Latino Professionals For America (ALPFA)	Riverside, CA
Director of Professional Development (OC)	April 2017 – Present
• Networked with 100+ professionals and recruiters in various firms and organizations, bringing out 10+ companies including Deloitte, KPMG, Vanguard, and Kohl's for our Professional Networking Dinner with 120+ attendees. • Proposed research and data from campus statistics and Career Center representatives to Big Four advisors on findings, improvements, and implementations to bring the 10+ accounting firms to UC Riverside to recruit 250+ students.	
Associated Students of UC Riverside (Student Government)	Riverside, CA
Elected CHASS Senator	May 2016 – June 2017
• Managed a \$1.6 million budget consisting of student funds from CHASS college, organizing meetings to allocate \$10,000+ weekly to student organizations and initiatives involving grants, cultural events, and university projects. • Advised students and faculty during my own office hours for 10 hours per week while implementing impactful projects such as the Academic Test Bank System, ASUCR Career Fair, and various legislation for 22,000+ students.	
SKILLS, INTERESTS, & CURRENT INVOLVEMENT	
• Skills: Microsoft Office (Excel, PowerPoint) CRM (Zendesk, Confluence, JIRA) G-Suite SQL (Class) • Interests: Strategy & Operations, Analytics, Career Consulting, Speaking at Universities, Basketball, Working Out.	

Resources & Moving Forward



Other Resources

- Stay updated by joining the discord and LinkedIn Groups
- Stay updated in with the news:
 - <https://zicklin.baruch.cuny.edu/current/graduate-advisement-hub/campus-resources/free-subscriptions-for-zicklin-students/>
- Join Academic/Professional Groups:
 - American Finance Association
 - Society for Industrial and Applied Mathematics (SIAM) (NYC Chapter)
 - CFA Society (NYC Chapter)
 - Society of Quantitative Analysts (SQA) (NYC Chapter)
 - International Association for Quantitative Finance (IAQF) (NYC Chapter)
 - Thalesians (NYC Chapter)
 - QWAFAEW (NYC Chapter)
 - Other NY based Mathematics/Finance/Comp Sci groups

FQE Club Resources



COMPUSTAT

Bloomberg

yahoo!
finance

S&P Global
Market Intelligence

THE AMERICAN FINANCE ASSOCIATION
Publishers of The Journal of FINANCE



& More

Baruch Project Resources

- <https://www.janestreet.com/puzzles>
- WRDs:
 - <https://remote.Baruch.cuny.edu/login?url=https://wrds-www.Wharton.upenn.edu/connect/>
 - <https://guides.newman.Baruch.cuny.edu/accessFinData/WRDS>
- Cap IQ:
 - <https://guides.newman.baruch.cuny.edu/AccessFinData/SandPDatabases>
- Bloomberg (Bloomberg Market Concepts/Terminal):
 - <https://guides.newman.baruch.cuny.edu/AccessFinData/BloombergProfessional>
- Factset
 - <https://guides.newman.baruch.cuny.edu/AccessFinData/FactSet>
- LSEG Workplace
 - <https://guides.newman.baruch.cuny.edu/AccessFinData/RefinitivWorkspace>

Q&A

Lets Jump into Resume Review